

Part VIII: the predict-the-output work

3 October 2026 — what was built, what was deliberately not, and the one decision I would like from you.

Why this happened

You asked for the interview material to be ground down "from very basic to very detailed," with brain-racking questions of the `int("25", 4)` kind.

Measured before starting: Part VIII held **565 coded questions** across its banks, and **thirteen of the fourteen banks had no predict-the-output section at all**. The only one was Chapter 72's §72.2, which is why it became the template.

(An earlier note of mine put the total at 619. That figure came from a different counting method and I could not reproduce it, so the numbers here are the ones I can show: distinct question codes appearing as a question heading or a rapid-fire row, counted from the repository before and after. 565 before, 767 now.)

You read §71.11 and approved its depth. This is that standard applied across the rest.

What was built

Ten new sections. 202 new questions. Part VIII goes from 565 coded questions to 767.

Ch	Bank	New section	Core	Rapid	Was → Now
70	Excel, Sheets, VBA & BI	70.9 What the grid does behind the number	6	10	71 → 87
71	SQL	71.11 Predict the output	16	16	76 → 108
72A	Data Structures & Algorithms	72A.10 Predict the output	16	16	30 → 62
73	Statistics & Experimentation	73.8 Arithmetic that quietly goes wrong	8	11	37 → 56
74	Machine Learning	74.8 What the library did that you did not ask for	10	10	36 → 56
75	Product Sense & Metrics	75.7 Metric arithmetic	7	8	31 → 46
77	Data Engineering	77.8 What happens to rows on the way in	9	12	35 → 56
78	Automation & Integration	78.7 Schedules, retries and defaults	7	10	28 → 45
79	GenAI, LLM & MLOps	79.9 The arithmetic behind an AI system	6	10	39 → 55
80	Architecture & Leadership	80.7 The arithmetic of a design	6	8	29 → 43

Every section follows §71.11's shape: a memory hook, a one-line answer, the code, its real output, a three-tier table, likely follow-ups, a red flag and a *Learn it in* pointer. A script (`tools/review-export/check_bank.py`) checks that every question has all five required parts, that no question code is duplicated or skipped, that code fences balance and that no table has a ragged row. **All ten new sections pass it completely — 0 of 202 new questions is missing any part.**

One thing that check turned up, worth passing on. Run across the *whole* of each chapter rather than just the new section, it flags **37 pre-existing questions** that lack one or more of those parts. Most are not defects: the full walk-throughs and design cases (Q73-036, Q74-035, Q75-030, Q77-033, Q79-038, Q80-022 and their neighbours) are deliberately a different format — a talked- through scenario rather than a question with a tier table — and several Chapter 71 queries treat the query itself as the answer. A handful may be genuine gaps, Q71-060 among them. I have not touched any of them; the list is reproducible by running the script without `--section`.

Every answer was run. Nothing was typed from memory.

Three banks I did not touch, and why

This is the part I would most like you to look at, because it is a judgement call.

Chapter 76A — Data Analyst & Data Scientist. This bank is entirely about career positioning and defending your own work: "walk me through a project you're proud of", "defend this number on your resume", "why this role and not that one". There is no technical or numeric content for a predict-the-output question to attach to. Adding one would be padding.

Chapter 81 — Behavioural, HR & Offer Conversations. Same reason for the behavioural part. The one numeric topic — CTC against in-hand pay — **is already covered**, in Q81-024, with a worked ₹6 lakh breakup table. A second treatment would duplicate it.

Chapter 82 — Take-Home Assignments & Mock Interviews. It has no coded questions at all. It is a different format — briefs and rubrics — and should stay that way.

The one decision I would like from you

Chapter 76B, the Business Analyst bank, is the borderline case. Its content is requirements, BRDs, user stories, BPMN and UAT — so "predict the output" does not fit it either.

But there is a genuine equivalent for a BA, and it is arguably the BA's core skill: **given a written requirement, what is ambiguous, and what will break when it reaches a developer?** A section of "here is a requirement as a stakeholder wrote it — find the three things that are undefined" would drill exactly that.

It is a different format from the other ten, so I have not built it unilaterally. **Say the word and I will.**

What the work kept finding

Worth knowing, because it changed what got built:

These banks were better covered than the headline number suggested. Chapter 73 already simulated peeking *and* the rare-disease Bayes problem. Chapter 74 already had leakage in four questions. Chapter 70 was already the largest bank in the book. So rather than hitting a quota, each section was aimed at what its chapter genuinely lacked — which is why Chapter 71 got 32 new questions and Chapter 73 got 19.

Four questions were cut for duplicating existing ones, in Chapter 73. Keeping the peeking question would have printed 26.2% beside the existing Q73-025's 16.3% — both correct for their own number of looks, and together a contradiction a reader would have had to untangle for nothing.

About a dozen drafted questions were wrong, and running them is the only reason they are not in the book. A few examples:

- `RANK` on the tied months gives 4, 4, 4, not the 1, 1, 1 I had written.
- `257 is 257 is True` in Python 3.12 — the compiler folds the literals, so the obvious way to write that question teaches the opposite of the truth.
- Removing the even numbers from `[1, 2, 3, 4, 5, 6]` while looping gives the *right* answer, so the famous skip bug is invisible on that data.
- `sum([0.1]*10) == 1.0` is `True`, and a million rupee amounts summed as floats matched the exact integer total — so the float-money question was cut rather than dressed up.
- pandas already treats `'N/A'` and `'NULL'` as missing, so that trap was backwards; the real one is a legitimate `NA` meaning *North America* being destroyed on the way in.
- Parquet was **slower** than CSV for a full read of a 300,000-row file. The question says so, and puts the real advantages where they belong.

Each of those is recorded in the chapter's changelog rather than quietly corrected.

What is honestly outstanding

1. **Chapter 71's §71.11 has still not been run on PostgreSQL 16 and MySQL 8.4.** Neither is installed here. Its outputs are labelled *Run (riverstone_2025)* rather than *Verified*, and four questions are marked **Dialect split**. 2. **Chapter 70's §70.9 was not run in Excel**, for the same reason. Four questions carry a **Check in Excel** mark naming the documented behaviour and why it is worth confirming. 3. **Book 4's visual check should be redone.** It has grown substantially.

Neither of the first two is hidden in a changelog — both are stated in the chapter's own at-a-glance box, so a reader is told before they rely on it.